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Automating aCRF PDF Page Number Population in SDTM Specifications Using SAS

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Abstract

Populating the PDF Page Reference column in the Study Data Tabulation Model (SDTM) specification VariableMetadata tab is a critical but time-consuming task in clinical trial data submissions. For each SDTM variable collected on a Case Report Form (CRF), the corresponding annotated CRF (aCRF) page number must be identified and entered manually. For a typical Phase 2/3 study, the aCRF can span hundreds of pages across dozens of SDTM domains and hundreds of variables — yet only a fraction of rows in the VariableMetadata tab may have PDF Page Reference values populated at any given time, illustrating the scale of the manual effort required.

This paper describes a three-step automated process using SAS to populate the PDF Page Reference column efficiently and accurately. First, the aCRF PDF is opened in Adobe Acrobat, the "Create Comment Summary" feature is used to generate a comment summary document, and the text is copied and saved as a plain text file. Second, analyze the text file and exclude irrelevant text by get_page.sas program and extract SDTM variable names and record the page number(s) on which each variable appears. Third, the SDTM specification Excel file is read directly using the SAS LIBNAME XLSX engine, the page number dataset is merged with the VariableMetadata sheet on variable name, and the updated specification is written back to Excel in place.

The result is a significant reduction in manual effort, improved accuracy, and a reproducible process applicable across studies. Importantly, the process targets only CRF-origin variables — the variables that are directly collected on the CRF and therefore annotated on the aCRF.

Introduction

The Study Data Tabulation Model (SDTM), developed and maintained by the Clinical Data Interchange Standards Consortium (CDISC), provides a standardized framework for organizing clinical trial data submitted to regulatory agencies such as the U.S. Food and Drug Administration (FDA).

SDTM organizes data into domains — structured datasets representing specific types of observations such as adverse events (AE), laboratory results (LB), vital signs (VS), and demographics (DM). Each domain is described in a specification document that defines the variables, their attributes, derivation rules, and traceability back to the source data.

A key component of SDTM traceability is the annotated Case Report Form (aCRF). The aCRF is a PDF version of the blank CRF with text box annotations overlaid on each page, mapping CRF fields to their corresponding SDTM variable names. Regulatory agencies require that SDTM submissions include an aCRF and that the SDTM variable metadata include PDF page references pointing to the aCRF pages where each variable is collected (FDA, 2022; CDISC, 2022). The PDF Page Reference column in the VariableMetadata tab serves this purpose — but only for variables whose values are directly collected on the CRF. Variables that are derived, assigned, or sourced from the protocol do not require page references.

For a typical Phase 2/3 clinical study, the aCRF can span hundreds of pages across more than 20 SDTM domains. Manually identifying the aCRF page number for each CRF-collected variable — by opening the PDF, searching for the annotation, and entering the page number into the Excel file — is tedious, error-prone, and time-consuming. The objective of this paper is to describe an automated, reproducible three-step process using SAS to populate the PDF Page Reference column, leveraging Adobe Acrobat's comment summary functionality, a SAS analyzing program, and the SAS LIBNAME XLSX engine.

Background

The Annotated CRF (aCRF)

An annotated Case Report Form (aCRF) is a PDF version of the blank CRF with text box annotations overlaid on each page. These annotations are created using PDF annotation tools such as Adobe Acrobat and contain SDTM variable names, domain labels, and derivation notes that map each CRF field to its corresponding SDTM representation. The aCRF is a required component of regulatory submissions.

The annotation format follows a consistent pattern. Each annotated page contains one or more text box annotations identifying an SDTM variable name (e.g., BRTHDTC, AGE, SEX), a domain label (e.g., "DM = Demographics"), or a derivation note (e.g., "VSORRES when VSTESTCD = HEIGHT"). Some annotations include supplemental qualifier references such as "RACEOR in SUPPDM." For example, the Demographics (DM)

domain page contains annotations including BRTHDTC, AGE, AGEU, SEX, ETHNIC, and RACE; Vital Signs (VS) pages contain VSDTC, VSORRES, and VSSTAT across multiple visit-specific pages.

The SDTM Specification File

The SDTM specification is an Excel workbook containing multiple tabs that collectively define the structure and content of all SDTM datasets for a study. The tabs include VariableMetadata, DatasetMetadata, VLM (Value Level Metadata), WhereClauses, and others.

Table 1: SDTM Specification file

	A	B	C	D	E		J	N	O	P	Q	R	S
1	Dataset Name	Parent Domain	Split Parent	Relative Order	Variable Name	ngth	Length	Transformation Rule Details for Clinical Data Integration Tool	Display Forma	Controlled Terminology	Evaluator	Origin	PDF Page Reference
53	DM	DM	DM	100	STUDYID	15		COPY AS IS from STUDYID				Protocol	
54	DM	DM	DM	200	DOMAIN	2		Hardcode DOMAIN = DM(GLOBAL CONVE		DOMAIN		Derived	
55	DM	DM	DM	200	DOMAIN	2		Hardcode DOMAIN = DM		DOMAIN		Assigned	
56	DM	DM	DM	300	USUBJID	25		COPY AS IS from USUBJID				Derived	
57	DM	DM	DM	400	SUBJID	12		COPY AS IS from SUBJID				CRF	2
58	DM	DM	DM	500	RFSTDTC	19			ISO 8601			Derived	
59	DM	DM	DM	600	RFENDTC	19			ISO 8601			Derived	
60	DM	DM	DM	700	RFXSTDTC	19			ISO 8601			Derived	
61	DM	DM	DM	800	RFXENDTC	19			ISO 8601			Derived	
62	DM	DM	DM	900	RFICDTC	19		Equal to DS.DSSTDTC where	ISO 8601			CRF	6
63	DM	DM	DM	1000	RFPENDTC	19			ISO 8601			Derived	
64	DM	DM	DM	1100	DTHDTC	19		COPY AS IS DS.DSSTDTC where DSDECOD=DEATH;	ISO 8601			Assigned	
65	DM	DM	DM	1200	DTHFL	1		If DS.DSDECOD=DEATH then		NY_01		Derived	
66	DM	DM	DM	1300	SITEID	7		COPY AS IS from SITEID				Assigned	
67	DM	DM	DM	1400	INVID	100						Derived	
68	DM	DM	DM	1500	INVNAM	200						Derived	
69	DM	DM	DM	1600	BRTHDTC	19		Concatenate BRTHDAT and	ISO 8601			CRF	5
70	DM	DM	DM	1700	AGE	8		COPY AS IS from AGE				CRF	5
71	DM	DM	DM	1800	AGEU	6		COPY based on codelist from AGEU_STD		AGEU		CRF	5
72	DM	DM	DM	1900	SEX	1		COPY based on codelist from SEX_STD		SEX		CRF	5
	DM	DM	DM	2000	RACE	50		(1) Derived from RWHITE, RBLACK, RASIAN, RAIAN,		RACE		CRF	5

DatasetMetadata | VariableMetadata | SupqualVariables | CompMethod | VLM | WhereClauses | SharedComments | ExternalDictionary | StudyInfo | SuppDocs

The VariableMetadata tab is the primary focus of this paper. Each row represents one SDTM variable within a specific domain. The columns include Dataset Name, Variable Name, Variable Label, Type, Length, Origin, Role, Core, Derivation, and — critically — column S, "PDF Page Reference." This column should contain the aCRF page number(s) where each CRF-collected variable is annotated, formatted as space-separated integers (e.g., "266 268 270" for a variable appearing on three pages). In a typical study, many CRF-collected variables lack page references at the time of analysis, representing the target population for the automated process described in this paper.

The Adobe Acrobat Comment Export

Adobe Acrobat provides a "Create Comment Summary" feature that generates a structured document listing all comments including annotations from a PDF file.

Methodology

Overview of the Three-Step Process

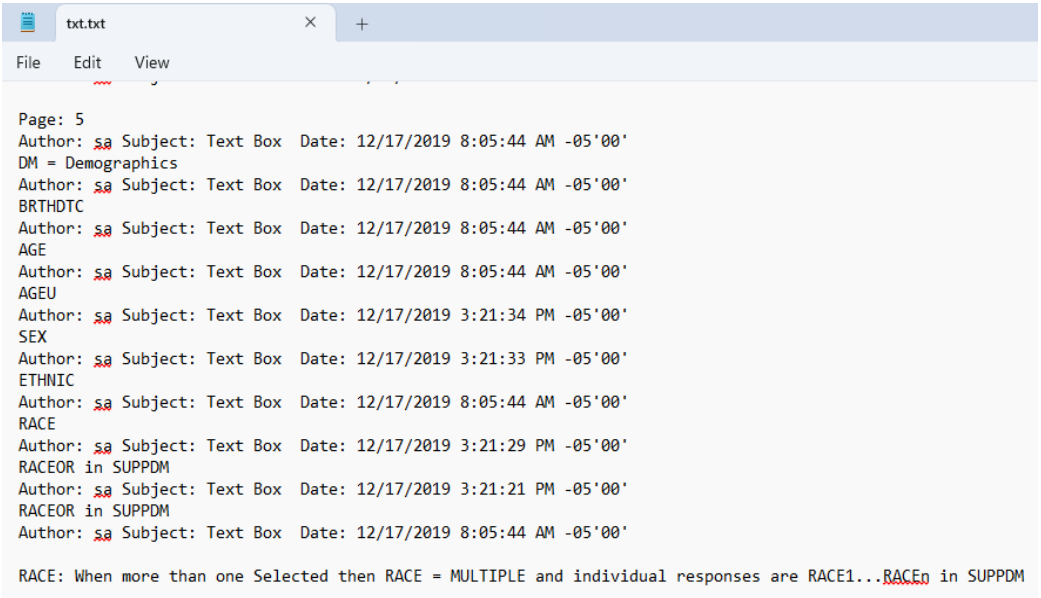
The automated process consists of three sequential steps: (1) extracting annotation comments from the aCRF PDF into a text file, (2) analyzing the pattern and run SAS program to keep information of page numbers for each annotated variable, and (3) using the SAS LIBNAME XLSX engine to read the SDTM specification Excel file, merge the page numbers into the VariableMetadata sheet, and write the updated data back to Excel in place.

Step 1: Extracting aCRF Annotations to a Text File

The first step exports all text box annotations from the aCRF PDF to a plain text file using Adobe Acrobat:

1. Open the aCRF PDF in Adobe Acrobat.
2. Click the **3 dots** (more options) menu in Adobe Acrobat.
3. Choose **"Create Comment Summary"**.
4. In the dialog, select the **"Comments only"** option.
5. In the resulting summary document, select all text and copy it.
6. Paste the copied text into a new .txt file and save it as **txt.txt**.

Table 2: Comments file extracted from aCRF



The resulting file contains all annotations organized by page number. Each annotation entry includes the page number, author, subject type (Text Box), timestamp, and the annotation content. Note that some pages contain only "NOT SUBMITTED" or "[NOT SUBMITTED]" annotations, indicating that the CRF data from those pages is not submitted in the SDTM datasets.

Step 2: Analyze the Text file and exclude irrelevant text by get_page.sas Program

The get_page.sas program reads the text files and extracts the page number for each annotated SDTM variable. The program operates as following (See **Appendix 1** for the details of get_page.sas):

First analyze pattern of the lines that contains no annotated Variable information, then flag the lines will be excluded from variable extraction. Lines are flagged for exclusion when: the line contains "Author: sa Subject: Text Box Date:", the line is exactly "Author: sa Subject: Text Box", the first word is "Date:", the line is "[NOT SUBMITTED]" or "NOT SUBMITTED", or the 3rd–5th characters are " = " (domain labels such as "DM = Demographics"). A PROC FREQ of excluded lines and a PROC PRINT of remaining lines are produced for review.

The page number of each variable is retained using a RETAIN statement. "Page: N" lines are detected and the page number is extracted using SCAN and INPUT functions. Lines where the first word (var) is not all uppercase are flagged (flg=1) as non-variable lines. For lines containing "=" in the variable content, only the part before "=" is kept (e.g., "DSDECOD = RANDOMIZED" → var = "DSDECOD"). Flagged lines are filtered out. The resulting dataset is sorted by var and page with duplicates removed using NODUPKEY, then all page numbers per variable are concatenated into a space-separated string (pagec). The variable 'VISIT' is excluded from the final output.

The output dataset (_temp4) contains two columns: var (the SDTM variable name as it appears in the aCRF annotations) and pagec (space-separated page numbers where the variable is annotated).

Step 3: Merging Page Numbers into the SDTM Specification

The final step reads the SDTM specification Excel file directly using the SAS LIBNAME XLSX engine, merges the page numbers from _temp4 into the VariableMetadata sheet, performs QC checks, and writes the updated data back to Excel in place(see the last part of get_page.sas in **Appendix 1** for the details).

Several important considerations apply to this step:

LIBNAME XLSX engine. Rather than using PROC IMPORT and PROC EXPORT, this step uses the SAS LIBNAME XLSX engine to read and write the Excel file directly. The `options validvarname=any;` statement is required to handle column names containing spaces and special characters, such as 'Variable Name'n and 'PDF Page Reference'n.

Sort order preservation. A sort order variable (myord=_n_) is added before sorting by variable name for the merge. After the merge, the dataset is sorted back to the original row order by myord before writing to Excel.

Updating Origin and PDF Page Reference. When a match is found (b=1), the Origin column is set to 'CRF' and the 'PDF Page Reference'n column is populated with the concatenated page numbers from pagec.

QC check (_check flag). The *check=1 flag is set for rows that already had a PDF Page Reference value but were NOT found in the new aCRF annotation data. A PROC FREQ on these rows filtered to STUDY*FLAG='Y' provides a targeted QC report identifying potential discrepancies — for example, variables from a previous aCRF version or variables requiring manual review.

Writing back to Excel. The final DATA step writes the updated dataset directly back to sdtmspec.VARIABLEMETADATA, overwriting the VariableMetadata sheet in place. Other sheets in the workbook are not affected.

Results and Discussion

The three-step automated process successfully populated PDF Page Reference values for the majority of CRF-collected variables that had aCRF annotations. Table 1 illustrates sample results across several SDTM domains.

Table 1: Sample PDF Page Reference Values Populated by the Automated Process

Domain	Variable Name	PDF Page Reference (Before)	PDF Page Reference (After)
AE	AEREFID	266 268 270	266 268 270
AE	AESEV	(empty)	266
AE	AESER	(empty)	266 269
DM	BRTHDTC	(empty)	5
DM	AGE	(empty)	5
DM	SEX	(empty)	5
VS	VSDTC	(empty)	32
VS	VSORRES	(empty)	32 35 37 39 41 43 45
LB	LBDTC	(empty)	68 72 105 114 152 156

EX	EXSTDTC	(empty)	245
DS	DSDECOD	(empty)	1 6 7 303 304
IE	IECAT	(empty)	8

The results demonstrate several important characteristics of the automated process. Many SDTM variables are annotated on multiple aCRF pages because the same CRF form is repeated across multiple visits — for example, VSORRES (Vital Signs Result) appears on seven pages corresponding to different vital sign measurements, and LBDTC (Laboratory Date/Time) appears on six pages corresponding to different laboratory panels. The get_page program correctly consolidates all page numbers for each variable into a single space-separated string.

The analyzing line pattern approach provides important quality control capabilities. The flag for exclusion helps verify which lines are being excluded — the programmer reviews this to ensure no valid variable names are accidentally filtered out before proceeding to the extraction pass. The _check=1 flag in Step 3 identifies variables that already had a PDF Page Reference in the specification but were NOT found in the new aCRF annotation data. These may be variables from a previous aCRF version or variables requiring manual review. The PROC FREQ on *check=1 rows filtered to STUDY*FLAG='Y' provides a targeted QC report focused on variables relevant to the current study.

Derived, Assigned, and Protocol variables are not annotated on the aCRF and therefore do not receive page references from the automated process — which is the correct behavior per CDISC guidelines. The primary limitation of the process is that it relies on the quality and completeness of the aCRF annotations. If a variable's annotation text does not match the expected SDTM variable name format, the variable will not be matched and will require manual review.

Conclusion

This paper has described a three-step automated process using SAS to populate the PDF Page Reference column in the SDTM specification VariableMetadata tab. The process — extract, parse, and merge — replaces a time-consuming and error-prone manual task with a reproducible, programmatic approach. For a typical study with a 300+ page aCRF and thousands of rows in the VariableMetadata tab, the automated process reduces the effort from potentially many hours of manual work to a process that can be completed in minutes once the initial setup is in place.

The key benefits are: (1) significant time savings for large studies; (2) improved accuracy through consistent parsing logic; (3) built-in QC through review approach and the _check flag; (4) reproducibility across studies; and (5) correct targeting of only CRF-origin variables, in alignment with CDISC and FDA submission requirements. Future enhancements could include programmatic handling of "Refer Annotations from Page X-Y" cross-references, integration with study setup workflows, and adaptation for aCRF annotation formats from other PDF tools or electronic data capture systems.

Appendix 1: the SAS program get_page.sas

```
/*To read in txt file, go get page numbers for each annotated
variable Zemin Zeng 2023-09-09*/

filename dsn "C:\Users\I0257844\OneDrive -
Sanofi\HomeDirMig\sas\get_page_number\20260728\Summary of Comments
on CRF Version.txt" lrecl=32767;

data _temp;
  length  col1 $ 2000;
  infile dsn truncover dsd ;
  input col1 ;
run;

*to retain page number;
data _temp2;
  set _temp;
  retain page;
  col1=strip(col1); lnth=length(col1);
  if index (col1, 'Page:') then page=
  input(strip(scan(col1, 2, ':') ), best.);/*store Page number*/
  if strip(col1)='' then delete;
  var=strip(scan(col1, 1, ''));
  if lnth>=5 then sub=substr(col1, 3, 3);

  if index(var, '=') then var=scan(var, 1, '=');

  if index (col1, 'Author: ')  or col1='[NOT SUBMITTED]' or
      col1='NOT SUBMITTED' or var='Date:' or  sub=' = '
      or var ne upcase(var)
  then flg=1;
run;

/*to review these lines to be excluded*/
proc freq data=_temp2;
  tables flg*var*col1/list missing;
run;

proc print data =_temp2 noobs;
  var col1;
```

```
  where flg ne 1;
run;

data _temp3; /*exclude these irrelevant lines*/
  set _temp2;
  where flg ne 1;
run;

proc sort data=_temp3 nodupkey;
  by var page;
run;

data _temp4;
  set _temp3;
  by var;
  length pagec $200;
  retain pagec;
  if first.var then pagec=strip(put(page, best8.));
  else pagec=strip(pagec) || ' ' || strip(put(page, best.));
  if last.var;
  if  var not in ('VISIT' );
  keep var pagec;
run;

/*Excel book --SDTM specs*/
options validvarname=any;

libname sdtmspec xlsx "C:\Users\I0257844\OneDrive -
Sanofi\HomeDirMig\sas\get_page_number\SDTM_spec.xlsx";

data varmeta;
  set sdtmspec.VARIABLEMETADATA;
  myord=_n_;
  length var $2000;
  var='Variable Name'n;
run;

proc sort data=varmeta;
  by var;
run;

data varmetau (drop=pagec);
  merge varmeta(in=a) _temp4(in=b);
  by var;
  if a;
  if b then do;
    Origin='CRF';
    'PDF Page Reference'n =pagec;
  end;
  if ('PDF Page Reference'n ne '') and not b then _check=1;
run;

proc sort data=varmetau out=varmetau2(drop=myord var);
  by myord;
run;
```

```
data sdtmspec.VARIABLEMETADATA;
  set varmetau2;
run;
```

References

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